Lecture 21

Jacobi functions and the pendulum

Objectives. *The pendulum period is an elliptic integral. Its motion is described by the inverse integral, a Jacobi function. We construct the real Jacobi functions by an ordinary differential equation, prove their periods and verify the pendulum formula with its initial conditions. The limit of these librations is a separatrix motion with infinite period.*

By the end of this lecture you will be able to:

- Use the Jacobi invariants to prove existence for all real time.
- Recover the quarter-period from the complete elliptic integral.
- Verify the positive-crossing pendulum solution and its least period.
- Prove compact-time convergence to the separatrix as the modulus tends to one.

21.1 A real differential equation

Fix a *modulus* k with $0 < k < 1$. Its *complementary modulus* is $k' = \sqrt{1 - k^2}$; this prime is not a derivative. The argument of the Jacobi functions is $u \in \mathbb{R}$. Define three real functions by the initial-value problem

$$s' = cd, \qquad c' = -sd, \qquad d' = -k^2 sc, \qquad (s, c, d)(0) = (0, 1, 1). \tag{21.1}$$

Primes on s, c and d mean derivatives with respect to u. The component $d(u)$ is a function, distinct from the polynomial degree used in Lecture 19. The vector field is polynomial, so local existence and uniqueness follow from Theorem 2.1.

Proposition 21.1. *The solution exists for all $u \in \mathbb{R}$ and satisfies*

$$s^2 + c^2 = 1, \qquad d^2 + k^2 s^2 = 1, \qquad k' \le d \le 1. \tag{21.2}$$

Proof. Differentiate the first two expressions using (21.1):

$$(s^2 + c^2)' = 2scd - 2csd = 0, \qquad (d^2 + k^2 s^2)' = -2k^2 dsc + 2k^2 scd = 0.$$

Their initial values are one. Thus $|s|, |c| \le 1$ and $d^2 \ge 1 - k^2 > 0$. Since $d(0) = 1$, continuity forces $d > 0$ and the stated bounds follow. The trajectory stays in a compact cube in $\mathbb{R}^3$. At a finite maximal endpoint this contradicts compact escape, Theorem 2.4. The same argument applies in both time directions. ∎

The *Jacobi functions* are these components:

$$\operatorname{sn}(u, k) = s(u), \qquad \operatorname{cn}(u, k) = c(u), \qquad \operatorname{dn}(u, k) = d(u).$$

Lawden, pp. 36–37, gives the same ODE and initial values.[1] At $k = 0$, the same ODE gives $(\sin u, \cos u, 1)$. For $k > 0$, the third component also changes with u.

21.2 The first quarter-period

Recall the complete elliptic integral

$$K(k) = \int_0^{\pi/2} \frac{\mathrm{d}v}{\sqrt{1 - k^2 \sin^2 v}}.$$

For $k < 1$ it is finite. Its relation to the ODE requires showing when the solution first reaches $s = 1$.

Proposition 21.2. *On $0 < u < K(k)$ the functions s and c are positive and s is strictly increasing. At the endpoint,*

$$(s, c, d)(K(k)) = (1, 0, k').$$

Proof. Define the strictly increasing continuous function

$$J(r) = \int_0^r \frac{\mathrm{d}q}{\sqrt{(1 - q^2)(1 - k^2 q^2)}}, \qquad 0 \le r \le 1.$$

The integrable square-root singularity at $r = 1$ causes no divergence, since $1 - k^2 q^2 \ge k'^2$. Substitution $q = \sin v$ gives $J(1) = K(k)$. Let s be the inverse of J on $[0, K(k)]$, and set $c = \sqrt{1 - s^2}$ and $d = \sqrt{1 - k^2 s^2}$ there. On the open interval, inverse differentiation gives $s' = cd$. Differentiating the other two roots gives $c' = -sd$ and $d' = -k^2 sc$. They have the required initial values and positive signs. The right-hand sides extend continuously to both endpoints, so their integral equations extend there as well. For $-K(k) < u < 0$, define

$$s(u) = -s(-u), \qquad c(u) = c(-u), \qquad d(u) = d(-u).$$

These reflections satisfy the ODE on $-K(k) < u < 0$. Their left derivative at zero is $(1, 0, 0)$, equal to the right derivative. Thus the constructed triple solves the initial-value problem on the open interval $(-K(k), K(k))$. By Theorem 2.1 it agrees there with the global ODE solution, and continuity identifies the endpoint. Its values at $K(k)$ are $(1, 0, k')$ as claimed. ∎

This proves an inverse-integral description on the first quarter, rather than presupposing a periodic solution. For instance, the argument where $\operatorname{sn}(u, k) = 1/2$ in that quarter is exactly $J(1/2)$. Other quarters follow from uniqueness.

21.3 Reflections and real periods

Theorem 21.3. *The least positive real periods of* sn *and* cn *are $4K(k)$. The function* dn *has period $2K(k)$.*

[1] His earlier definition uses theta quotients. We instead use the real ODE specified here, so no result from Lecture 22 is required. His k is the modulus; software often takes the parameter $m = k^2$.

Proof. For this proof abbreviate $K = K(k)$. The reflected triple $(s(K-v), -c(K-v), d(K-v))$ solves the same ODE in v as $(s(K+v), c(K+v), d(K+v))$. The derivative of its first component is $-cd = (-c)d$. The derivative of its second component is $c' = -sd$. Here the original components on the right are evaluated at $K - v$. The third equation follows in the same way, since changing the sign of c cancels time reversal. At $v = 0$ both triples are $(1, 0, k')$. Uniqueness gives

$$s(K+v) = s(K-v), \quad c(K+v) = -c(K-v), \quad d(K+v) = d(K-v).$$

In particular the state at $2K$ is $(0, -1, 1)$. The transformation $(s, c, d) \mapsto (-s, -c, d)$ preserves the ODE. Translate the solution by $2K$ and use uniqueness at zero to obtain

$$s(u+2K) = -s(u), \qquad c(u+2K) = -c(u), \qquad d(u+2K) = d(u).$$

Applying this twice gives a $4K$ period for s and c.

The first-quarter positivity and the reflection formulas give $s > 0$ on $(0, 2K)$ and $s < 0$ on $(2K, 4K)$. Its only zeros in $[0, 4K)$ are 0 and $2K$, and $s'(2K) = -1$ differs from $s'(0) = 1$. Any positive period smaller than $4K$ would take zero to another zero with the same derivative, which is impossible. Similarly, the same formulas give $c < 1$ on $(0, 4K)$ and $c(0) = 1$; thus c has no smaller positive period. ∎

The proof concerns real arguments. Complex poles, imaginary periods and the other Jacobi quotients are not needed for the pendulum.

21.4 The positive-crossing pendulum

In this lecture the pendulum angle is θ; Lecture 2 wrote it as x. This local change separates the angle from other coordinates. It does not denote the Heaviside function of Lecture 8 or the theta function $\theta(z \mid \tau)$ of Lecture 22. The positive constant ν_0 is the pendulum's frequency scale. The equation and positive-crossing initial state are

$$\ddot{\theta} = -\nu_0^2 \sin\theta, \qquad \theta(0) = 0, \qquad \dot{\theta}(0) = 2k\nu_0. \tag{21.3}$$

A *libration* is a periodic swing between two turning angles without a full rotation.

Theorem 21.4. *For $0 < k < 1$, the solution of (21.3) is*

$$\theta(t) = 2\arcsin\bigl(k\,\mathrm{sn}(\nu_0 t, k)\bigr). \tag{21.4}$$

Its turning amplitude is $\theta_{\max} = 2\arcsin k$, so $k = \sin(\theta_{\max}/2)$. Its least positive period is

$$T = \frac{4K(k)}{\nu_0}. \tag{21.5}$$

Proof. The real arcsine has values in $[-\pi/2, \pi/2]$. Since $|k\,\mathrm{sn}| \le k < 1$, the formula is smooth for all t. The second invariant and positivity of dn give $\cos(\theta/2) = \mathrm{dn}(\nu_0 t, k)$. Differentiate (21.4); the dn factor cancels its positive arcsine denominator, giving

$$\dot{\theta} = 2k\nu_0\,\mathrm{cn}(\nu_0 t, k), \qquad \ddot{\theta} = -2k\nu_0^2\,\mathrm{sn}(\nu_0 t, k)\,\mathrm{dn}(\nu_0 t, k) = -\nu_0^2 \sin\theta.$$

The initial values of sn and cn give both initial conditions. ODE uniqueness identifies this with the pendulum motion. The first-quarter endpoint gives the positive turning angle, and

the reflections give its negative. Since arcsine is one-to-one here, a period of the angle is exactly a period of sn after scaling time by ν_0. Theorem 21.3 proves the stated least period. ∎

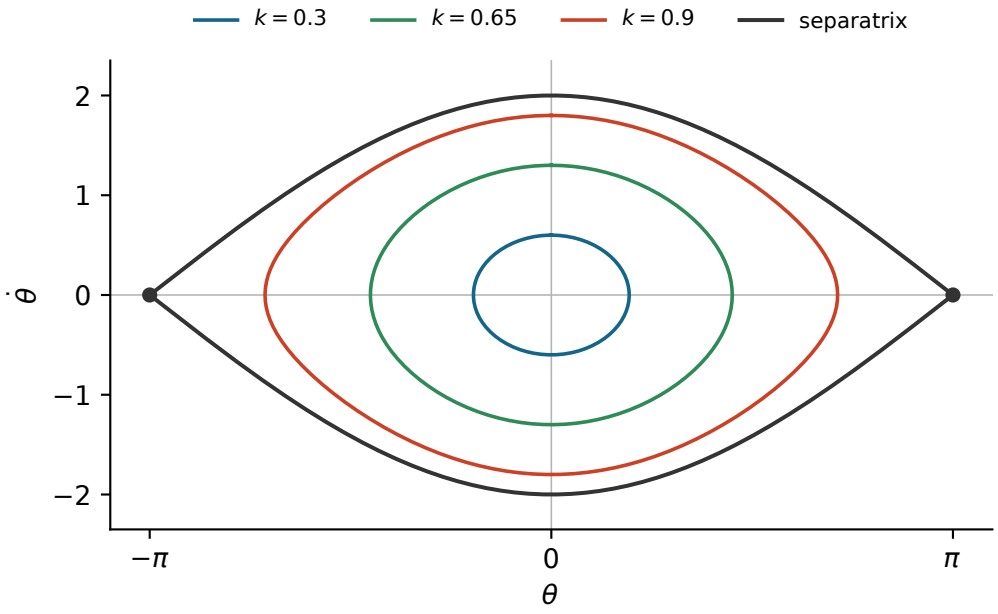
For example, if $k = 3/5$ and $\nu_0 = 2$, the initial angular velocity is $12/5$, and the exact period is $2K(3/5)$. The formula specifies the phase at the bottom with positive velocity; a different starting phase requires a time translation. Lawden, pp. 114–115, uses this same phase.[2]

21.5 The separatrix limit

The conserved energy per unit moment of inertia is

$$\mathcal{E} = \frac{1}{2}\dot\theta^2 + \nu_0^2(1 - \cos\theta).$$

Its derivative is $\dot\theta(\ddot\theta + \nu_0^2 \sin\theta) = 0$. The initial state gives $\mathcal{E} = 2k^2\nu_0^2$. The level $\mathcal{E} = 2\nu_0^2$ separates librations from rotations. A nonconstant motion on this boundary is called a *separatrix*. The states $\theta = \pm\pi$, $\dot\theta = 0$ are upright equilibria. The phase portrait displays the librations and this limiting energy curve.



For $\nu_0 = 1$, each closed curve is a libration at the indicated modulus. The separatrix approaches the upright equilibria at $\theta = \pm\pi$. The script uses exact energy curves, not a numerical ODE approximation.

Proposition 21.5. *As $k \uparrow 1$, the Jacobi triple converges uniformly on every compact real u interval to $(\tanh u, \operatorname{sech} u, \operatorname{sech} u)$. The corresponding pendulum angles converge uniformly on compact time intervals to*

$$\theta_{\mathrm{sep}}(t) = 2\arcsin(\tanh(\nu_0 t)) = 4\arctan e^{\nu_0 t} - \pi.$$

The periods of the librations tend to infinity.

[2] His frequency ω is our ν_0, and his swing amplitude α is our $\theta_{\max}$. No small-angle approximation is made.

Proof. Here $\operatorname{sech} u = 1/\cosh u$. The derivative identities $(\tanh u)' = \operatorname{sech}^2 u$ and $(\operatorname{sech} u)' = -\tanh u \operatorname{sech} u$ verify all three ODE equations at $k = 1$ and the initial values. Uniqueness identifies this as the endpoint solution.

For every finite t, $|\tanh(\nu_0 t)| < 1$, so the differentiation in Theorem 21.4 also applies at $k = 1$. It gives

$$\dot{\theta}_{\mathrm{sep}} = 2\nu_0 \operatorname{sech}(\nu_0 t),$$
$$\ddot{\theta}_{\mathrm{sep}} = -\nu_0^2 \sin\theta_{\mathrm{sep}}.$$

The initial angle is zero and the initial velocity is $2\nu_0$. The energy is therefore $\mathcal{E} = 2\nu_0^2$, the separating level defined above. This verifies that the endpoint angle is a separatrix motion of the same pendulum.

Convergence follows from a uniform parameter estimate. On the invariant cube $|s|, |c|, |d| \le 1$, the field has Lipschitz constant at most 3, uniformly for $0 \le k \le 1$. Its Jacobian has at most six nonzero entries of absolute value at most one, so its operator norm is at most $\sqrt{6} < 3$. At a fixed state the fields for parameters k and 1 differ in only the third component, by at most $1 - k^2 \le 2(1 - k)$. For modulus k, write the components as s_k, c_k and d_k. Write the error norm as

$$e_k(u) = \|(s_k, c_k, d_k)(u) - (s_1, c_1, d_1)(u)\|.$$

Subtract the integral equations and use the two field bounds. For $0 \le u \le U$ this gives

$$e_k(u) \le 2(1 - k)u + 3\int_0^u e_k(v)\,\mathrm{d}v.$$

The term proportional to u is not the constant term in the form of Grönwall proved in Lemma 2.2. Define the shifted error by

$$h_k(u) = e_k(u) + \frac{2(1 - k)}{3}.$$

The integral of the added constant absorbs that term, giving

$$h_k(u) \le \frac{2(1 - k)}{3} + 3\int_0^u h_k(v)\,\mathrm{d}v.$$

Applying Grönwall to this inequality gives

$$h_k(u) \le \frac{2(1 - k)}{3} e^{3u}.$$

Subtracting the constant shift gives the error estimate.

For $|u| \le U$ this gives

$$\|(s_k, c_k, d_k)(u) - (s_1, c_1, d_1)(u)\| \le \frac{2(1 - k)}{3}(e^{3U} - 1).$$

For negative u apply the same argument to the reversed vector field. Thus convergence is uniform on compact intervals. On a fixed compact interval the limiting $\tanh u$ stays strictly between -1 and 1. Multiplying s_k by k preserves its uniform convergence to this limit as $k \uparrow 1$, since the functions remain uniformly bounded. Continuity of arcsine therefore gives the angle convergence. The arctangent expression has the same derivative $2\nu_0 \operatorname{sech}(\nu_0 t)$ and the same value zero at $t = 0$, so it equals the arcsine expression.

Finally, the integrand defining $K(k)$ increases to $1/\cos v$ on $[0, \pi/2)$. Its integral is infinite, so monotone convergence (Theorem 3.2) gives $K(k) \to \infty$ and hence $T \to \infty$. ∎

The separatrix approaches $-\pi$ and π as $t \to -\infty$ and $t \to +\infty$, respectively. It is not a periodic orbit. The more precise logarithmic rate $K(k) = \ln(4/k') + o(1)$ is the sketch in Proposition 14.7; the qualitative divergence just proved does not depend on that asymptotic.

Summary. The real Jacobi ODE has a bounded global solution. Its first quarter inverts an elliptic integral, and uniqueness supplies the remaining quarters. The positive-crossing pendulum formula follows by differentiation.

- The libration amplitude satisfies $k = \sin(\theta_{\max}/2)$, and its least period is $4K(k)/\nu_0$ for $0 < k < 1$.
- As $k \uparrow 1$ the motion approaches the separatrix on compact time intervals while its period diverges.

References

Lawden, *Elliptic Functions and Applications*, pp. 24–31, 36–37 and 114–115. DLMF 22.13 records the derivative equations; DLMF 22.5 records the hyperbolic endpoint. The proofs here use the real ODE and its invariants.